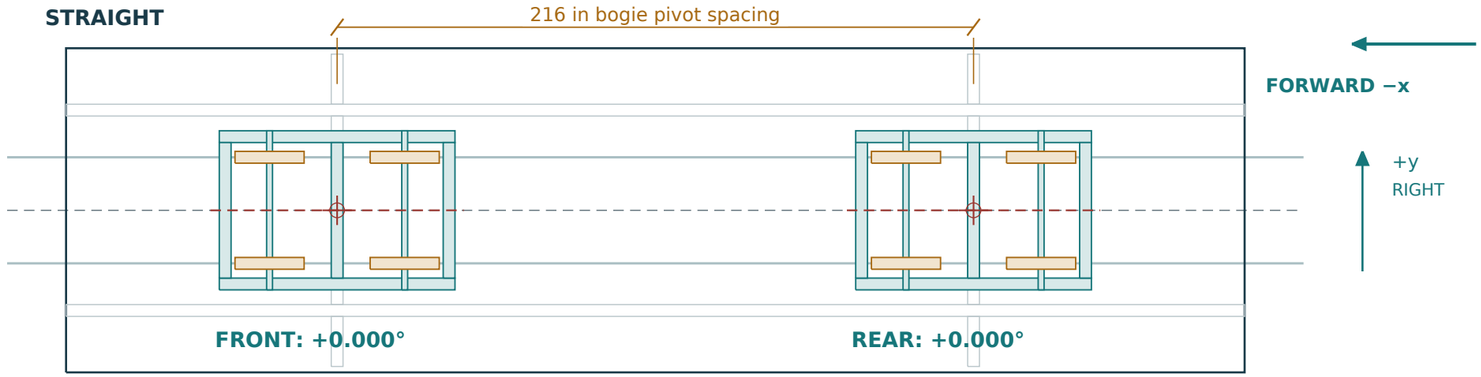
HOW THE TWO BOGIES TURN UNDER THE COACH

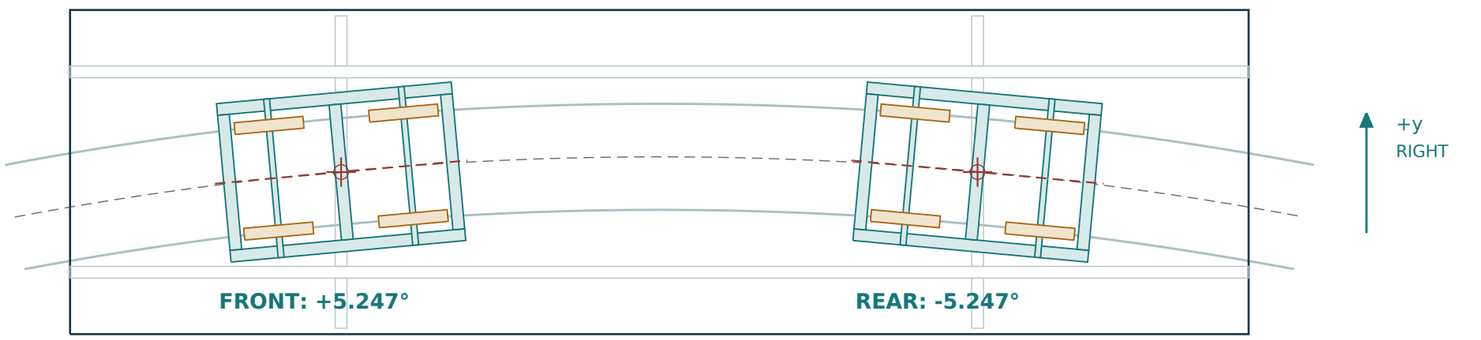
F1 32 in floor study only — full-motion clearance unresolved. All new undercarriage forms are DISPLAY / NOT PART geometry.

G1.1 BODY-FIXED PLANS / YAW CALCULATED FROM THE SHARED TURNING KERNEL

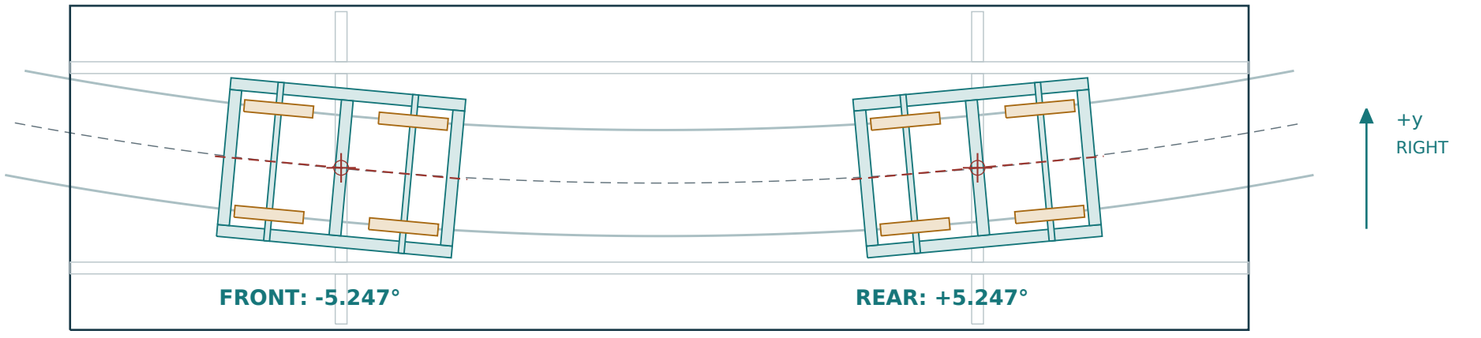
STRAIGHT



LEFT TURN / R = 30 m



RIGHT TURN / R = 30 m



G1.2 CALCULATED YAW / ANGLE OF ATTACK

R (m)	|bogie yaw|	|axle attack|
30	5.247°	1.113°
50	3.145°	0.668°
75	2.096°	0.445°

WHAT ROTATES

Each bogie carries two parallel axles and four wheel proxies. The whole bogie rotates about its body-fixed XY pivot. Front and rear yaws are equal and opposite in this symmetric steady-curve study.

THE WHEELSETS DO NOT STEER RADIALLY

Axle midpoints lie on the selected circle, but each axle remains fixed square to its bogie. Nonzero angle of attack remains; no wheel/rail force, flange contact or derailment calculation is shown.

COORDINATE / RADIUS DEFINITIONS

Forward is −x. +y is vehicle right. Positive curvature therefore means RIGHT. R is the axle-midpoint circle radius; bogie-pivot and body-centre radii are slightly smaller.

RETAINED GEOMETRIC PROXIES

1165 mm wheelbase; 596 mm nominal wheel cylinder; 101.6 mm width. 914.4 mm wheel-centreplane separation is a visual proxy, not an accepted back-to-back or wheel/rail profile.

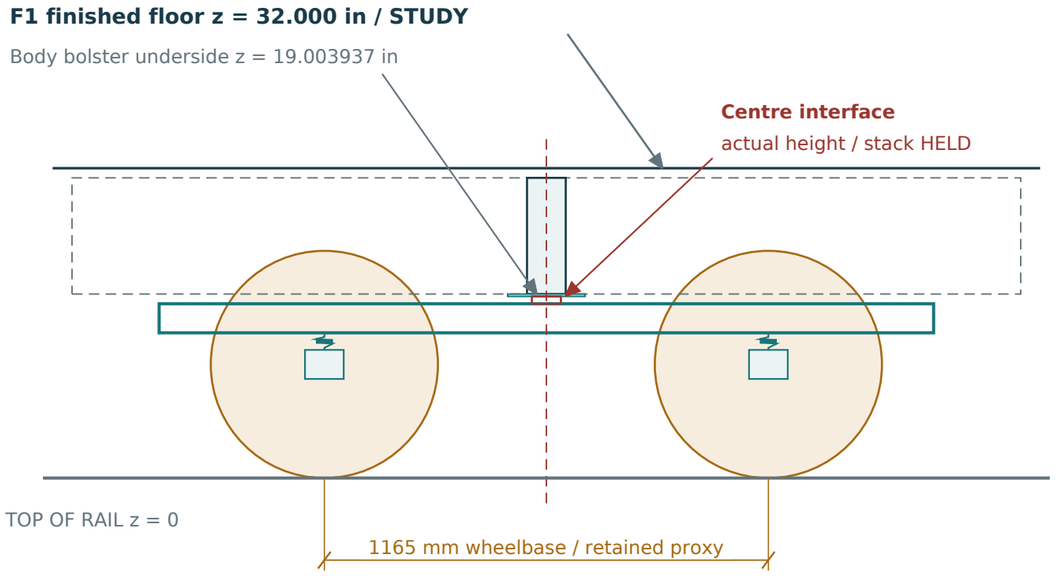
Body and display parts shown to plan scale. Track lines are reference guides; actual rail/profile, suspension, brakes and hardware remain HELD.

VERIFIED / source data CALCULATED / stated inputs STUDY / proposed geometry HELD / required before release

SUPPORT FUNCTIONS / SIDE AND TRANSVERSE INTERFACES

Displayed bogie forms explain assembly relationships. No kingpin, centre bearing, sidebearer or suspension part is selected.

G2.1 FRONT BOGIE SIDE PROJECTION / STRAIGHT

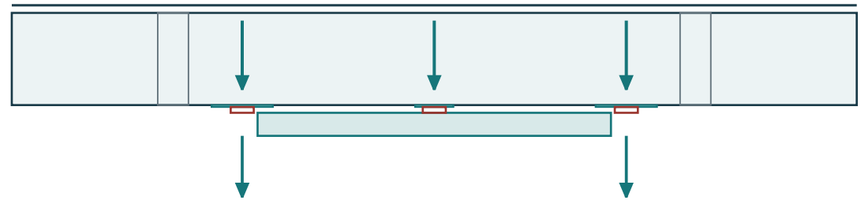


Sideframe, axlebox and spring shapes are DISPLAY ONLY; projected parts are at different transverse planes.

G2.2 TRANSVERSE SUPPORT FUNCTION / x = 45

CENTRE AND/OR SIDE SUPPORTS

Vertical load sharing is undefined. Side support gaps / preload, roll couple, uplift retention and local body reinforcement require one engineered interface.



To suspension → axleboxes → wheelsets → rail

Centre marker supplies an axis, not an approved pin or bearing.

Wheelsets lie ahead and behind this section. No wheels are projected here, so section overlap cannot be mistaken for an actual fit test.

G2.3 EXPLODED FUNCTIONAL ASSEMBLY / NOT A PARTS LIST OR FABRICATION DETAIL

BODY STRUCTURE
Retained F1 bolster; local joint / reinforcement HELD

→

BODY–BOGIE INTERFACE
Centre and/or side supports; locator, retention and stops HELD

→

BOGIE FRAME
One engineered braked assembly; actual frame / mounts HELD

→

PRIMARY SUSPENSION
Springs, damping and axlebox guidance; travel / rates HELD

→

WHEELSETS / RAIL
Wheel profile, bearings and rail geometry coordinated together

VERTICAL

Body support allocation must distinguish sprung body force from total rail reaction, including actual bogie self-mass.

DRAFT / BRAKING / LATERAL

Define force paths through the locating and retaining joints; no pin, plate, weld or bolt size is assigned here.

ROLL / LIFT / MOTION

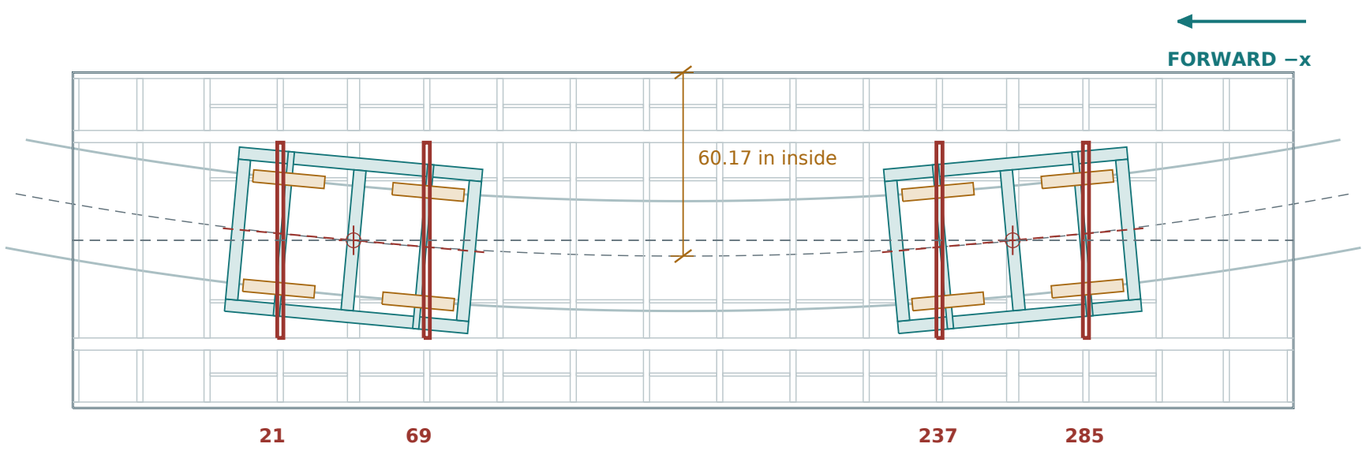
Return sidebearer gap or preload, allowable uplift, yaw stops, pitch/roll travel and combined motion limits.

VERIFIED / source data CALCULATED / stated inputs STUDY / proposed geometry HELD / required before release

CURVE SWEPT SPACE / RETAINED WHEEL–FRAME CONFLICTS

Rigid-body geometry is an input to clearance engineering; it does not establish a route envelope, minimum safe radius or hardware fit.

G3.1 RIGHT R30 PLAN / FRAME AND WHEEL PROXIES



Red transverse nodes: x = 21, 69, 237, 285 in. P3 bounding overlap persists at all four.

Track guide / 914.4 mm centreplane proxy is not the accepted wheel-to-rail interface.

G3.2 FRAME-ONLY STEADY-CURVE SPACE

R m	Inward	Inside	Outside
30	5.17	60.17	65.97
50	3.10	58.10	61.78
75	2.06	57.06	59.58

Offsets in inches from the axle-midpoint track circle.

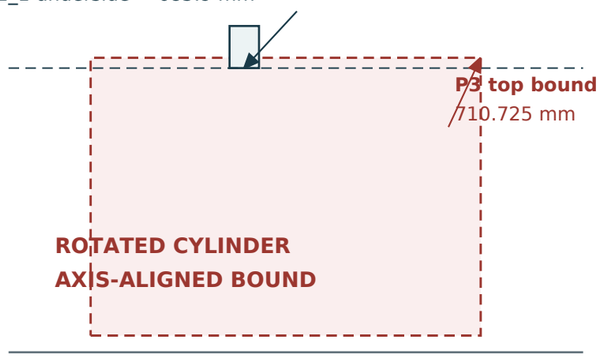
The 400 × 110 in rectangle includes the retained end platforms. Add projections beyond that rectangle, suspension motion, body roll, construction tolerances, track condition and the approved clearance margin.

92 in frame overhang beyond each pivot. Coupling articulation, stairs, guards, transitions and reverse curves need separate checks.

G3.3 P3 BOUND / T21_1

Side envelope; not an actual wheel shape.

T21_1 underside = 685.9 mm



Negative bound = possible overlap; not confirmed solid collision.

G3.4 F1 VERTICAL SCREENS / ALL OEM FIT RESULTS UNRESOLVED

Case	Radial / heave mm / roll	Gap mm	Method
P0	0 / 0.0 / 0°	+89.90	Exact cylinder
P1	0 / 25.4 / 0°	+64.50	Exact cylinder
P2	25 / 25.4 / 2°	+20.13	Bounding box
P3	35 / 50.8 / 3°	-24.82	Bounding box

Phase 3G repeats the nominal cylinder/F1 steel and equipment-reservation screens at straight and left/right R30, R50 and R75. No negative gaps occur in those sampled rest / +25.4 mm heave states. Historical 3F boxes are retained; new air equipment is not allocated. Actual bogie frames, axleboxes, springs, chambers, hoses and combined motion are not cleared.

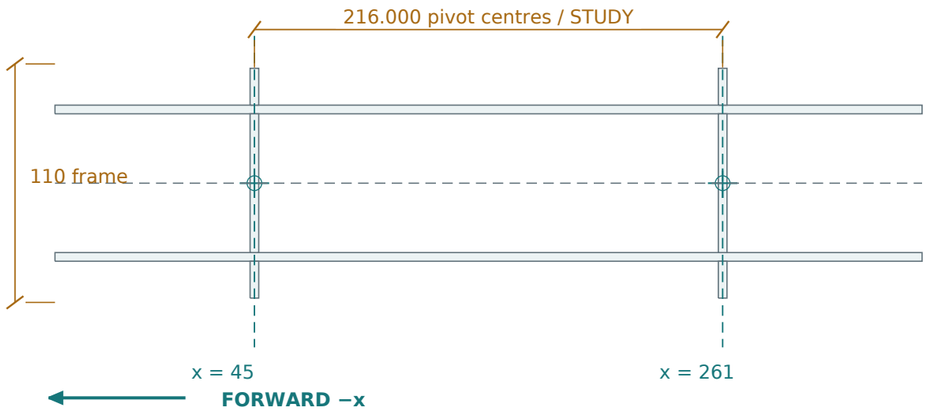
VERIFIED / source data CALCULATED / stated inputs STUDY / proposed geometry HELD / required before release

OEM RETURN DATUM / QUALIFICATION WORK PACKAGES

One complete braked-bogie design authority; controlled evidence must close the body, track and brake interfaces.

G4.1 COMMON DATUM RETURN / DIMENSIONS IN INCHES

Body reference: x = 0 front inside wall; y = 0 centreline; z = 0 top of rail.



Pivot symbols define XY datums only; actual support Z is HELD.

RETURN SECTION AT EACH BOGIE

OEM support top + attachment stack + accepted relative-motion reserve must fit the body underside. Do not allocate bearing force from the centre symbol alone.

F1 floor z = 32.000 study

Steel bearing plane z = 31.003937

12 in bolster underside z = 19.003937 (482.7 mm)

Actual loaded / unloaded support height: _____

Support coordinates, wheel wear and travel: _____

G4.2 CONTROLLED RETURNS / ALL ACTUAL VALUES HELD

RETURN	REQUIRED CONTROLLED EVIDENCE
GR01	Exact configuration, gauge, designer and supply scope
GR02	Loaded/unloaded support heights, stack and coordinates
GR03	Body reactions, self-mass, axle/wheel load split
GR04	Vertical, draft, lateral, roll and lift-retention paths
GR05	Suspension travel, stiffness, damping, stops and limits
GR06	STEP + full moving and maintenance-removal envelopes
GR07	Actual body curve / reverse curve / twist / speed limits
GR08	Wheel profile, back-to-back, rail and turnout compatibility
GR09	Air chamber force/stroke/mount, rail brake and trainline failures
GR10	Engineered mounts, materials, joints and production QA
GR11	Jacking, safe retention, access, recovery and spares
GR12	Accepted calculations, test protocol and acceptance plan

Each return needs document / revision / actual value / acceptance / owner.

Supplier inquiry candidates: Severn Lamb (complete 914 mm reference), Alan Keef (documented bespoke bogies), Western Train (custom-gauge coaches; air brakes). No separate-bogie offer, project fit, quote or lead time verified.

CR-G01: air service + spring parking requested; hardware / integration HELD.

G4.3 RELEASE SEQUENCE / NO TEST LOADS OR ACCEPTANCE LIMITS ASSIGNED HERE

1 DEFINE
OEM returns + responsible engineer

→

2 INTEGRATE
Body / track / brake interface design

→

3 VERIFY FIT
Unoccupied travel, rotation and removal

→

4 QUALIFY
Accepted loaded, brake and track trials

→

5 RELEASE
Evidence / inspection / operating limits

Retain 3E E03–E05 / 3F verification objectives; revise air-brake procedures and criteria for CR-G01. Full-consist / authority gates remain.

VERIFIED / source data CALCULATED / stated inputs STUDY / proposed geometry HELD / required before release